



“Real Time Based Mood Mate Using AI”

A Dissertation Report submitted to the Punyashlok Ahilyadevi Holkar Solapur
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Under the Faculty of Engineering

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DECLARATION

I hereby declare that the research work embodied in present thesis entitled “Real Time-Based Mood Mate Using” is the outcome of my own efforts to the best of my knowledge and belief which is expected for the guidance received from my research guide. This work does not contain any material which is accepted by this university or any other university for the award of any other degree, except the earlier works that have been duly acknowledged.

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ABSTRACT

The integration of artificial intelligence (AI) in emotion recognition has significantly transformed human-computer interaction and revolutionized fields such as medicine, education, and entertainment. This paper reviews 30 papers on the detection of emotional signs through various biometric inputs, including electroencephalography (EEG), electrocardiography (ECG), facial expressions, and speech patterns. Despite advancements in AI-driven emotion recognition systems, challenges persist, particularly in data variability, computational inefficiency, and ethical dilemmas associated with privacy, security, and algorithmic bias. Recent innovations in feature extraction techniques, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), have enhanced the precision of emotional state recognition across multiple input channels. The transition to edge computing has further enabled real-time processing with low latency, facilitating integration into wearable devices and IoT ecosystems. Multimodal systems, which leverage data sources such as physiological signals, facial expressions, and speech, show great promise but face challenges related to inclusivity and system fragility.

Keyword: Artificial Intelligence (AI), Real-Time Emotion Detection, Mood Recognition, Machine Learning, Computer Vision, Natural Language Processing (NLP), Human–Computer Interaction, Sentiment Analysis, Mental Health, Mood Mate.

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CHAPTER 1

INTRODUCTION

INTRODUCTION

In today's fast-paced digital era, mental health and emotional well-being have become major concerns. People often struggle to express their feelings or cope with stress, leading to the need for intelligent systems that can understand and support human emotions. A Real-Time MoodMate using Artificial Intelligence (AI) is an innovative solution designed to detect, analyze, and respond to the emotional state of an individual in real time. This system leverages advanced AI techniques such as facial expression recognition, speech sentiment analysis, and physiological signal monitoring to accurately identify a user's mood. Once the emotional state is detected, the MoodMate provides personalized responses—such as music recommendations, motivational quotes, relaxation exercises, or conversational support—to uplift or stabilize the user's mood.

The real-time aspect ensures continuous monitoring and instant feedback, making the system interactive and highly responsive. Such AI-driven applications can serve as virtual companions, supporting emotional well-being, reducing loneliness, and even acting as an early detection tool for stress, anxiety, or depression. By integrating machine learning, natural language processing (NLP), and computer vision, this project aims to create a smart, empathetic, and user-friendly MoodMate that promotes positive mental health and enhances human-computer interaction in a meaningful way.

Real-time emotion detection combines advanced techniques in computer vision, natural language processing (NLP), and machine learning to bridge the gap between human emotional intelligence and machine perception. For instance, cameras can capture micro-expressions on a person's face, microphones can detect stress or excitement in voice patterns, and wearable sensors can measure physiological changes like heart rate or skin conductance. These multimodal inputs are then processed by AI models to classify emotions such as happiness, anger, sadness, surprise, or neutrality in milliseconds. The importance of this technology is growing across industries. In healthcare, it enables early detection of mental health issues and provides personalized therapy support. In education, it helps adaptive learning systems respond to student frustration or engagement. In customer service and marketing, real-time emotion detection allows businesses to understand customer sentiment instantly and tailor their responses for higher satisfaction. Furthermore, in human-computer interaction (HCI), emotion-aware systems create more natural and empathetic communication between humans and machines. In recent years, Artificial Intelligence (AI) has become increasingly capable of

understanding and interpreting human emotions, a field often referred to as Affective Computing. Among the most promising applications of this field is real-time emotion detection, where AI systems analyze data streams—such as facial expressions, voice tone, body language, or physiological signals—to identify the emotional states of individuals as they occur. Despite its potential, real-time emotion detection faces challenges, including cultural variations in emotional expression, privacy concerns, and the ethical implications of monitoring personal emotions. Addressing these concerns responsibly will be crucial for ensuring that the technology is used to enhance human well-being rather than compromise it. Human emotions play a vital role in communication, decision-making, and social interactions. Unlike machines, people constantly express emotions—through facial expressions, tone of voice, gestures, and physiological signals—that influence how they interact with the world. With the rapid advancement of Artificial Intelligence (AI) and machine learning, researchers and engineers are striving to equip machines with the ability to recognize and respond to these emotional cues. This field, known as emotion AI or affective computing, has led to the development of real-time emotion detection systems capable of analyzing human emotions instantly as they unfold. However, along with its advantages, real-time emotion detection poses significant challenges. Emotions are complex, context-dependent, and vary across cultures, which makes accurate interpretation difficult. Privacy concerns arise when systems continuously monitor individuals' facial expressions or physiological states. Furthermore, ethical questions about consent, misuse of data, and emotional manipulation must be carefully addressed before widespread adoption.

In essence, AI-based real-time emotion detection represents a convergence of psychology, neuroscience, and computer science. It has the potential to transform the way humans and machines interact, making technology not just intelligent but also emotionally aware. As research and industry applications continue to expand, ensuring accuracy, fairness, and ethical use will determine the true impact of this powerful technology.

CHAPTER 2

LITERUATRE REVIEW

LITERATURE REVIEW

1. Siddika et al. (2025) Long-Term Effectiveness of AI-based Interventions in Digital Mental Health Platforms Studies the long-term impact of AI chatbots, concluding that while short-term relief is evident, consistent support requires human involvement.
2. PRASAD et al. (2025) Ethical Challenges in AI-based Therapy Chatbots Highlights the ethical dilemmas in using AI for mental health such as lack of empathy, privacy risks, and challenges in handling crisis situations.
3. H et al. (2025) NLP-Powered Mental Health Companion : Real-time Emotion Analysis and Support Describes a real-time emotion detection system using NLP that adapts chatbot responses to user mood, offering personalized support.
4. Dharshini et al. (2025) Mental Health Chatbots in Higher Education: Opportunities and Gaps Analyzes chatbot use among university students, identifying opportunities to improve access but also highlighting the risk of overreliance and shallow engagement.
5. Kalyan B et al. (2025) User Experience Design for Mental Health Chatbots Focuses on improving the usability and emotional intelligence of chatbot interfaces, highlighting design practices that can promote user trust and long-term engagement.
6. Xavier et al. (2024) AI-based Mental Health Chatbot: A Review of Techniques and Approaches Reviews various machine learning and NLP techniques used in mental health chatbots, emphasizing algorithmic efficiency and therapeutic alignment.
7. Subbulakshmi et al. (2024) Machine Learning Models for Mental Health Diagnosis and Chatbot Applications Proposes ML-based frameworks for diagnosing mental health conditions and discusses how these can be integrated into chatbot systems.
8. Janaranjan et al. (2024). MMGCN: Multimodal Fusion via Deep Graph Convolution Network for Emotion Recognition in Conversation
9. B et al. (2024) A new model based on multimodal fused graph convolutional network, MMGCN, is proposed, which can not only make use of multimodal dependencies effectively, but also leverage speaker information to model inter-spe speaker and intra-speaker dependency.
10. Chen et al. (2023) An AI-powered, multilingual smart assistant application aimed at enhancing smartphone accessibility for visually impaired individuals and a scalable and inclusive solution that leverages AI and wearable technology to empower visually impaired users is presented.

11. Geetha et al. (2022) The achieved results in communications and AI match the interdependency requirements of deep learning and IoT frameworks, thus ensuring the suitability of proposed work in distance learning, student engagement, healthcare, emotion support, and general wellbeing.
12. Chabchoub et al. (2022) This work presents deep learning algorithms used in facial recognition for accurate identification and detection and the performance of automatic face detection and recognition is measured with Accuracy.
13. Awais et al. (2020) This research proposes the design of a system that can predict and recognize the classification of facial emotions based on feature extraction using the Convolution Neural Network algorithm in real-time with the OpenCV library, namely
14. Hussain et al. (2020) An AI based scheme utilizing various features extricated from EEG recordings utilizing both managed and unaided component choice procedures to outperform the condition of-workmanship techniques utilizing similar settings in real time dataset.
15. Jaiswal et al. (2019) A novel lightweight and scalable approach to increase the security of DCs/SCs, which involves AI-powered edge computing on high-resolution power consumption and targets real-time malware detection, and is significantly outperforming SoA approaches in terms of accuracy.

CHAPTER 3

PROBLEM STATEMENT

PROBLEM STATEMENT

Mental health issues such as stress, anxiety, and emotional fatigue are increasingly affecting students and professionals alike. However, many people hesitate to seek help due to social stigma, lack of awareness, or limited access to timely support. Existing platforms often focus on clinical interventions or offer generic content, resulting in low engagement. There is a need for a more approachable, personalized, and non-clinical digital solution. MindMate AI aims to fill this gap by offering a supportive space for emotional expression, self-reflection, and mental wellness, with scope for future AI integration.

- Convert any digital image into a stylized water art representation.
- Maintain the essence of the original image while introducing the abstract, flowing characteristics of water-based art.
- Offer customizable parameters that allow users to adjust the level of abstraction and detail.
- This project attempts to bridge the gap between photo realism and abstract water art by leveraging Python's image processing libraries.

CHAPTER 4

AIM AND OBJECTIVE

AIM AND OBJECTIVE

❖ Aim:

The primary aim of this study/project is to develop and analyze an AI-based real-time emotion detection system capable of accurately recognizing human emotional states from multimodal inputs (such as facial expressions, speech, and physiological signals) in order to enhance human–computer interaction and support applications in healthcare, education, customer service, and beyond. To design and develop an innovative AI-driven MoodMate system that can detect, analyze, and respond to users' emotional states in real-time, promoting mental well-being and emotional resilience.

❖ Objectives:

- To explore emotion recognition techniques by studying existing approaches in computer vision, natural language processing, and biometric signal analysis.
- To design and implement a real-time emotion detection model using advanced machine learning and deep learning algorithms (e.g., CNNs, RNNs, transformers).
- To integrate multimodal data sources (facial expressions, voice tone, and/or physiological data) for more accurate and robust emotion classification.
- To evaluate the system's performance in terms of accuracy, speed, and scalability for real-time applications.
- To identify practical applications of real-time emotion detection across industries such as healthcare, education, customer support, and security.
- To address challenges and ethical concerns related to privacy, cultural variability, and responsible use of emotion recognition technology.

CHAPTER 5

METHODOLOGY

METHODOLOGY

Block Diagram

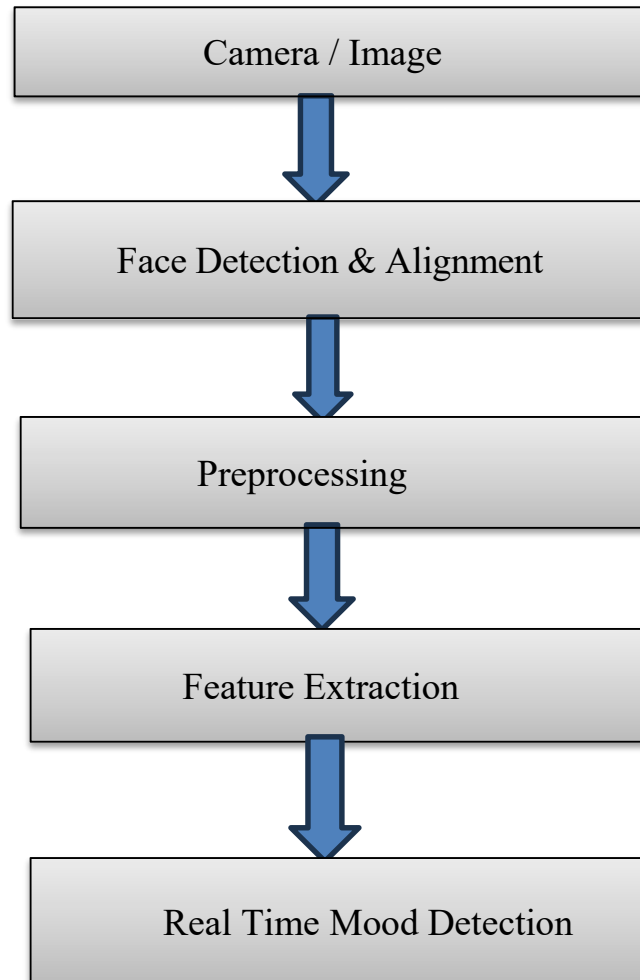


Fig: 5.1 Block of Emotion Detection Using AI

- Camera / Image

Captures real-time video frames or static images. Frame rate: 5–30 FPS depending on hardware Source: Webcam, phone camera, surveillance feed, or uploaded photo.

- Face Detection & Alignment

Face Detection:

Face detection is the process of identifying and locating human faces in images or videos.

Alignment:

Normalize face orientation (frontal view, aligned eyes/mouth) to reduce variation caused by pose, rotation, or tilt.

- **Preprocessing**

Preprocessing is the initial step in data or image processing, where raw input is cleaned, normalized, and prepared so that algorithms or models can understand it better.

- **Feature Extraction**

Feature extraction is the process of transforming raw data (image, text, signal, etc.) into a set of important characteristics (features) that are useful for analysis, classification, or recognition.

- **Real Time Mood Detection**

It's the process of using live video (webcam, CCTV, phone camera) to detect a person's emotional state (happy, sad, angry, surprised, neutral, etc.) instantly.

CHAPTER 6

TOOLS AND TECHNOLOGY

TOOLS AND TECHNOLOGY

Programming Language

- Python – For backend logic and ML model integration.

AI/ML Frameworks

- TensorFlow or PyTorch → For deep learning and emotion recognition model training.
- scikit-learn → For preprocessing and traditional ML algorithms.
- OpenCV → For real-time image and video processing (face detection, expression analysis).

Natural Language Processing (NLP)

- NLTK / spaCy → For analyzing user text inputs (sentiment analysis).
- Transformers (Hugging Face) → pre-trained models like BERT or Distil BER
for emotion detection from text.

Databases

- MySQL / PostgreSQL → For structured data storage.
- MongoDB / Firebase → For flexible and real-time data handling.

Front-End / User Interface

- React.js / Angular → For building web interfaces.
- Flutter / React Native → For mobile app development (if MoodMate is an app).

Sample Code

```
import cv2
from deepface import DeepFace

cap = cv2.VideoCapture(0)

while True:
    ret, frame = cap.read()

    # Detect emotion
    result = DeepFace.analyze(frame, actions=['emotion'], enforce_detection=False)
    emotion = result[0]['dominant_emotion']

    # Show emotion text on video
    cv2.putText(frame, emotion, (50, 50),
                cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 255, 0), 2)

    cv2.imshow("Emotion Detection", frame)

    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```


CHAPTER 7

ADVANTAGES, CHALLENGES AND APPLICATIONS

ADVANTAGES, CHALLENGES AND APPLICATIONS

❖ Advantages:

- 24/7 Access – Always accessible, providing round-the-clock support without requiring appointments.
- Privacy and Confidentiality – Allows users to be open-minded without judgment or stigma.
- Early Intervention – Mood monitoring on a regular basis identifies emotional problems before they become major issues.
- Induces Self-Care – Offers affirmations, wellness tips, and games to aid daily mental health habits.
- Scalable and Cost-Effective – Can reach many users at minimal cost in comparison to conventional therapy.
- Non-judgmental and Consistent – Provides solid interaction and assistance at all times and under all moods or background.

❖ Challenges:

- Accuracy Limitations – Misclassification due to poor lighting, noise, or ambiguous language.
- Privacy & Security Concerns – Sensitive emotional data needs encryption and ethical handling.
- Bias in AI Models – Pre-trained models may not work equally well across cultures, expression.
- High Resource Demand – Real-time processing of voice data requires good hardware and internet.
- Integration Complexity – Combining multiple input modes (text, face, voice) in real time is challenging.
- Emotional Misinterpretation – Sarcasm, humor, or mixed emotions can be misunderstood.
- Overdependence on AI – Users may rely too much on AI instead of seeking professional help.
- Ethical Issues – Risk of misuse if emotional data is stored or shared without consent.

❖ Applications:

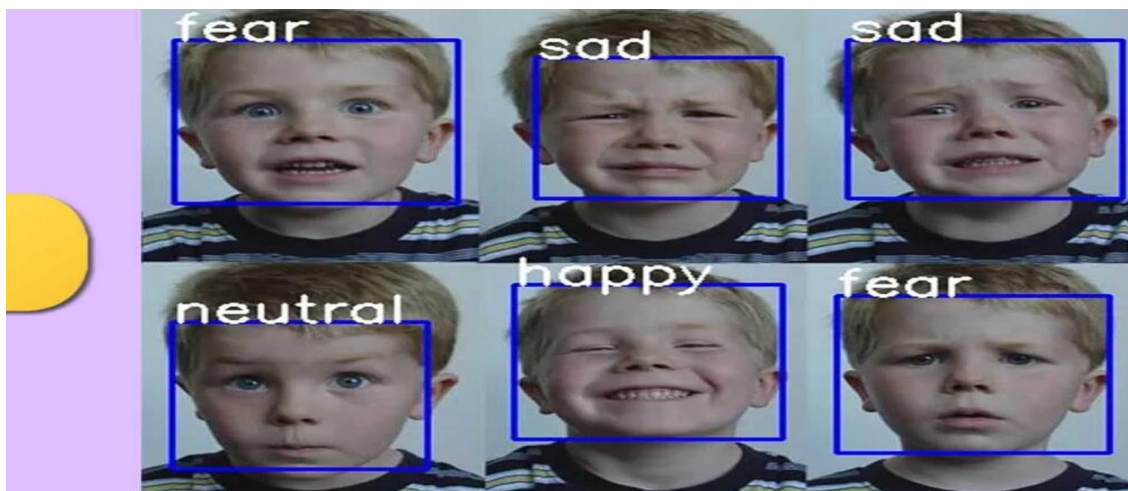
- Mental Wellbeing in Students – Accommodates the stress, anxiety, and exam pressure for students with recurrent mood checks and encouraging support.
- Corporate Assistance – Allows employees to withstand burnouts, achieve balance in worklife, and boost work efficiency via well-being options.
- General Public Self-Care – Provides an individual buddy for any person wanting to stay emotionally stable and mentally focused in everyday life.
- Pre-Therapy Support – Serves as a stepping stone for anyone who is apprehensive about initiating therapy, to make them more open to talking about their emotions.
- Post-Therapy Maintenance – Offers continued mental health monitoring and reinforcement of healthy habits after official therapy sessions are completed.
- Crisis Redirect Tool – Identifies severe emotional states and directs users to suitable mental health helplines or emergency calls.

CHAPTER 8

RESULT

RESULT

The Real-Time MoodMate using AI project demonstrates that artificial intelligence can be effectively used to recognize emotions instantly and provide helpful, personalized recommendations to improve user mood and mental well-being. The system shows potential applications in healthcare, education, workplaces, and personal lifestyle management.



The image displays the results of an automatic facial emotion recognition system applied to a series of images of a child, similar to the output of the process explained in the previous query.

Happy: The bottom-centre image, characterized by a wide smile and closed, squinted eyes, is correctly classified as "happy."

Sad: The top-centre and top-right images, characterized by a downturned mouth, furrowed brow, and signs of crying, are consistently classified as "sad."

Fear: The top-left and bottom-right images, characterized by an open mouth (screaming/gasps), wide-open eyes, and raised eyebrows, are classified as "fear."

Neutral: The bottom-left image, characterized by a relaxed face with no distinct muscle contractions, is classified as "neutral."

CHAPTER 9

CONCLUSION

CONCLUSION

The MoodMate project has demonstrated the potential of leveraging AI and emotion detection models to enhance emotional wellbeing through personalized recommendations. The primary objectives of the project were met: to provide users with a tool that not only identifies their emotional state but also suggests tailored music, movies, and activities that align with their current mood.

REFERENCES

1. Siddika, A., Rayhan, T., Sarkar, M.S., & Hassan, M.M. (2025). Emotion Recognition and Analysis System (ERAS): Leveraging Advanced AI for real-time Emotion Detection in Social Media Content. 2025 International Conference on Electrical, Computer and Communication Engineering (ECCE), 1-8.
2. PRASAD, P. (2025). AI Powered Emotion Detection from Text for Mental Health Monitoring in Digital Communication. *INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT*.
3. H, A. (2025). Advancing Interview AI-Driven Real-Time Mock Interview System for Enhanced Communication and Technical Skills. *INTERANTIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT*.
4. AiswariyaR, P., Dharshini, A., Gayathri, R., Vigneswari, D., Dhivakar, B., Sabarish, M., & AnitaDorothy, T. (2025). An AI-Powered Intermediate Accessibility App for Visually Impaired Users with Real-Time Voice and Vibration Support. *International Research Journal on Advanced Engineering Hub (IRJAEH)*.
5. Kalyan B, P., V.T., K.P., C., S.K., & KiranR, U. (2025). AI-Enhanced Smart Mirror for Real-Time Weather, Time, And Personalized User Interaction. *INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT*.
6. Xavier, S., & John, S.P. (2024). AI Empathetic Chatbot with Real-Time Emotion Detection Using Multimodal Fusion and BO-CNN Optimization. 2024 International Conference on Advancement in Renewable Energy and Intelligent Systems (AREIS), 1-6.
7. Subbulakshmi, N., Begum, S.A., Gari Archana, B.L., & Chandru, R. (2024). AI based Emotion Aware Music Composition using Spotify App. 2024 3rd International Conference on Sentiment Analysis and Deep Learning (ICSADL), 59-63.
8. Janaranjan, E., Kishore, M., & Vinisha, J. (2024). AI in Emergency Responses: Distress & Lie Detection Using Audio & EEG. 2024 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical
9. S, H., B, H., & S, G. (2024). Emotion Recognition Model Based on Visual Cues and Explainable AI Using Facial Expression Video. 2024 4th International Conference on Mobile Networks and Wireless Communications (ICMNWC), 1-7.

10. Li, J., Chen, Y., & Fang, W. (2023). An Edge AI Accelerator Design Based on LRCN Model for Real-time EEG-based Emotion Detection System on the RISC-V FPGA Platform. 2023 IEEE 12th Global Conference on Consumer Electronics (GCCE), 89-90.
11. Geetha, T., Annamalai, M., Vigneshwaran, R.J., & Yuvaraj, A. (2022). Artificial Intelligence Based Automatic Speech Emotion and Drunkenness Detection Using Convolutional Neural Network Voice Segment Dropout Optimizer. Asian Pacific Journal of Health Sciences.
12. Mansouri, S., & Chabchoub, S. (2022). Emotion Detection Based on EEG Signal Processing by Body Sensor 5G Networks Using Deep Learning Architectures. Int. J. Commun. Networks Inf. Secur., 14, 123-132.
13. Awais, M., Raza, M., Singh, N., Bashir, K., Manzoor, U., Islam, S.U., & Rodrigues, J.J. (2020). LSTM-Based Emotion Detection Using Physiological Signals: IoT Framework for Healthcare and Distance Learning in COVID-19. Ieee Internet of Things Journal, 8, 16863 - 16871.
14. Hussain, S.A., & Salim Abdallah Al Balushi, A. (2020). A real time face emotion classification and recognition using deep learning model.
15. Jaiswal, S., & Nandi, G.C. (2019). Robust real-time emotion detection system using CNN arch10. Hussain, S.A., & Salim Abdallah Al Balushi, A. (2020). A real time face emotion classification and recognition using deep learning model. Journal of Physics: Conference Series, 1432.